

## Daily Integral - Medium

$$\int_0^1 \int_{-1}^2 \int_{1001}^4 \dots \int_{1001}^4 dx \dots dx$$

This problem was generously submitted by HyphenGreaterThan on <https://dailyintegral.com>.

### Solution

Let us define  $I_n$  as the integral with the outermost upper bound as  $n$ . Then, we can see that our innermost integral is  $I_{1001}$ :

$$I_{1001} = \int_0^{1001} dx = 1001.$$

The next integral is:

$$I_{1000} = \int_{1001}^{1000} dx = -1.$$

The layer after that:

$$I_{999} = \int_{-1}^{999} dx = 1000.$$

And after that:

$$I_{998} = \int_{1000}^{998} dx = -2.$$

It appears that there is a pattern. As we saw, if  $n$  is even, then  $I_{n-2}$  is  $I_n - 1$ , starting from  $-1$  at  $I_{1000}$ . If  $n$  is odd, then  $I_{n-2}$  is  $I_n - 1$ , starting from  $1001$  at  $I_{1001}$ . From this, we can find that

$$I_n = \begin{cases} \frac{n-1002}{2} & \text{if } n \text{ is even} \\ \frac{1001+n}{2} & \text{if } n \text{ is odd} \end{cases}$$

Hence, since the outermost integral is  $I_1$ , and  $1$  is an odd number, we can find that

$$I_1 = \frac{1001+1}{2} = 501.$$